# Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

## 1 The decision

An indie studio's creative director picks the next game's two lead tags; if no pair passed the selection rule, they would pick a crowded one (table, last row).

## 2 Why a network

Counts show how common a pair is and how well it sells, not whether its tags belong to separate groups or which tag connects them. That needs the network.

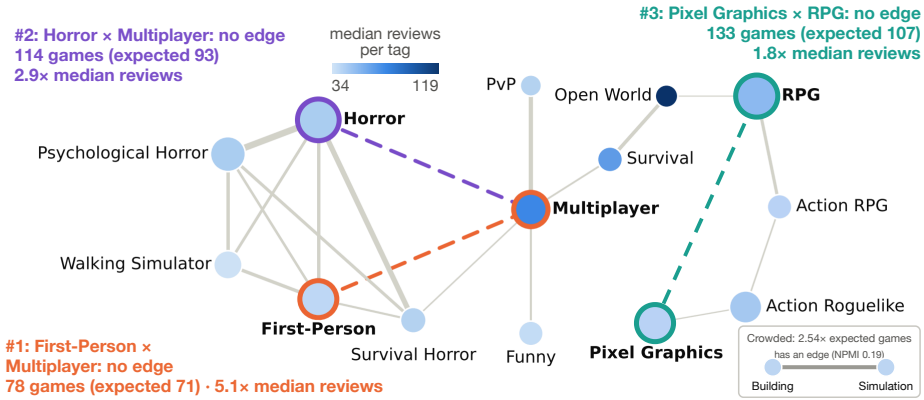
## 3 Data

| | |
|---|---|
| **Source** | S1 FronkonGames steam-games-dataset (Hugging Face); S2 live Steam store. |
| **Obtained** | S1 downloaded (Sep 2025). S2: 1,000 random S1 games (random seed 7202), scraped 2026-10-14. |
| **Legal** | S1 MIT licence. S2: robots.txt allows these pages; one request per 1.5 s; no personal data. |
| **Volume** | S1: 9,094 Indie games (2023–24, ≥10 reviews) + 2,076 from Jan–Jun 2025 as a test set. S2: 996. |
| **Check** | S1 and S2 agree: 87% overlap in top-5 tags; review counts correlate at 0.98. |
| **Bias** | Favours games that sold; reviews count buyers, not demand; late-2024 games had less time. |

## 4 Graph construction

| | |
|---|---|
| **Node** | A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph. |
| **Edge** | Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation. |
| **Resolution** | Co-op, Online/Local Co-Op and Local Multiplayer merged into Multiplayer (84% of Horror × Multiplayer games were also tagged co-op). |
| **Weight** | NPMI (co-occurrence above expected). Raw counts give 9 of the top 10 edges to the 5 most common tags; NPMI gives none. |
| **Window** | Top 5 tags per game, as the store page shows; with all 20, #1's advantage falls from 5.1× to 3.0×. |
| **Excluded** | "Indie"; Early Access, Free to Play, adult tags; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration. |
| **Pruning** | Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**. |

## 5 The picture




**First-Person and Multiplayer have no direct link; the shortest route between them goes through Survival Horror (2 links). Without it, the shortest route is 5 links.** 14 of 80 tags; size = games; line width = NPMI.

*Expected* = games that would carry both tags if developers picked them independently (A's games × B's games ÷ 9,094); *crowding* = games ÷ expected, above 1.5 = overdone; *median reviews* = the typical game's reviews. **Rule:** of pairs on ≥73 games, keep crowding ≤ 1.5 and ≤20% shared neighbours; rank by median reviews.

| Rank: pair | Games | Expected by chance | Crowding | Shared neighbours | Median reviews (typical indie: 40) |
|---|---|---|---|---|---|
| #1: **First-Person × Multiplayer** | 78 | 71 | 1.10× | 8% | 204 (5.1×) |
| #2: **Horror × Multiplayer** | 114 | 93 | 1.23× | 8% | 117 (2.9×) |
| #3: **Pixel Graphics × RPG** | 133 | 107 | 1.25× | 0% | 70 (1.8×) |
| Crowded (box): **Building × Simulation** | 74 | 29 | 2.54× | 8% | 136 (3.4×) |

## 6 Findings

- **#1** is on 78 games, close to the 71 expected, so it is not overdone. Yet its typical game gets 204 reviews, 5.1× the typical indie game ($p < 0.001$). Not because of horror: horror games get 1.0×.
- **Survival Horror is the only tag connecting** #1's two groups of neighbours: the one tag to watch.
- **Mostly robust:** top under all 8 pruning settings but only 59 of 100 random resamples; on unseen 2025 games, crowding 1.14, 2.4× typical reviews.

## 7 Recommended action

- Week 0: pitch a first-person multiplayer game.
- Weeks 1–6: prototype. Week 7: Coming Soon page with both tags.
- Every quarter: re-run and apply section 8.

## 8 What would change my mind

First-Person is spreading (27% of 2025 games). If #1's crowding passes 1.5, I switch to #2 while it stays below 1.5 (1.45 in 2025), then to #3.